

Hyperbolic geometry: A historical and conceptual overview

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Hyperbolic geometry is a non-Euclidean geometry that studies spaces with constant negative curvature. In the 18th century, several prominent mathematicians unsuccessfully attempted to eliminate the fifth postulate of Euclidean geometry (the parallel postulate) by deducing it from the first four postulates. However, it was in the 19th century that mathematicians recognized that merely assuming the first four axioms of Euclidean geometry could lead to an independent and consistent system of geometry.

This realization resulted in the discovery of hyperbolic geometry, which was first formalized by Nikolai Lobachevsky.

The groundbreaking work of William Thurston in the 1980s on the geometrization conjecture established that hyperbolic geometry is the most prevalent geometry in both two and three dimensions. As a result, hyperbolic geometry has become the backbone of modern low-dimensional topology owing to its close connections to various areas of mathematics, including differential geometry, dynamical systems and ergodic theory, geometric group theory, mapping class groups, and Teichmüller theory. In this talk, we will briefly trace the evolution of hyperbolic geometry and discuss some fascinating concepts in this field.